

Definition 1. Let $p \in Var$.

$$\phi ::= p \mid \phi \vee \phi \mid \neg\phi \mid B_i^{[l,u]} \phi \mid \phi_1 \mathbf{U} \phi_2$$

Definition 2. Let $T = (S, R, V)$ be a transition system, where S is the set of worlds, $R \subseteq S \times S$ and V a valuation. For each agent i , we define a probability transitions $P_i(s, s')$ such that $P_i(s, s') > 0 \implies sRs'$ and $\sum_{s'} P_i(s, s') = 1$.
The set of paths generated by the underlying transition systems is also called $\text{Paths}(T)$.

Definition 3. Let $\sigma = s_0s_1\dots s_n$ be a finite prefix. Let $C(\sigma) = \{\pi \in \text{Paths}(T) \mid \sigma \text{ is prefix of } \pi\}$. Then for each finite prefix and agent i we define the initial distribution of agent i to be $\mu_i^0(C(\sigma)) = \omega(s_0) \cdot P(s_0, s_1) \cdot \dots P_i(s_{n-1}, s_n)$.

Definition 4. Let π be a path and t a time. We define $C_i(\pi, t) = \{\pi' = s'_0s'_1\dots \in \text{Paths}(T) \mid s_k \sim_i s'_k \text{ for all } 0 \leq k \leq t\}$. Furthermore, given a set $A \subseteq \text{Paths}(T)$ the subjective posterior is

$$\mu_i^{\pi, t}(A) = \frac{\mu_i^0(A \cap C_i(\pi, t))}{\mu_i^0(C_i(\pi, t))}$$

Definition 5. Let T be a transition system, $\pi \in \text{Paths}(T)$ and t a timepoint. Then we have:

$$T, \pi, t \models p \quad \text{iff } \pi(t) \in V(p),$$

$$T, \pi, t \models B_i^{[l,u]} \phi \quad \text{iff } \mu_i^{\pi, t}(A_t(\phi)) \in [l, u] \text{ where } A_t(\phi) = \{\pi' \in \text{Paths}(T) \mid T, \pi', t \models \phi\}$$

Definition 6.